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DETAIL SPECIFICATION

MOUNT, RESILIENT: TYPE 5B5,000H

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers the type 5B5,000H resilient mount assembly. This mount has an assigned capacity of 3,500 to 5,000 pounds and a resonant frequency of 5±1 hertz over this load range.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-407 - Visual Inspection Guide for Rubber Molded Items

(Copies of this document are available online at http://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

803-1385709 - Mount, Resilient Type 5B5,000H and Auxiliary Snubbers

(Copies of this document are available from the applicable repositories listed in S0005-AE-PRO-010/EDM, which can be obtained online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/. Copies of this document may also be obtained from the Naval Ships Engineering Drawing Repository (NSEDR) online at https://199.208.213.105/webjedmics/index.jsp. To request an NSEDR account for drawing access, send an email to NNSY_JEDMICS_NSEDR_HELP_DESK@navy.mil.)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of this document is available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D395 - Standard Test Methods for Rubber Property – Compression Set

ASTM D412 - Standard Test Methods for Vulcanized Rubber and Thermoplastic Elastomers – Tension

ASTM D429 - Standard Test Methods for Rubber Property – Adhesion to Rigid Substrates

ASTM D471 - Standard Test Method for Rubber Property – Effect of Liquids

ASTM D573 - Standard Test Method for Rubber – Deterioration in an Air Oven

ASTM D792 - Standard Test Methods for Density and Specific Gravity (Relative Density) of Plastics by Displacement

ASTM D1005 - Standard Test Method for Measurement of Dry-Film Thickness of Organic Coatings Using Micrometers

ASTM D1149 - Standard Test Methods for Rubber Deterioration – Cracking in an Ozone Controlled Environment

ASTM D1229 - Standard Test Method for Rubber Property – Compression Set at Low Temperatures

ASTM D2240 - Standard Test Method for Rubber Property – Durometer Hardness

ASTM D2632 - Standard Test Method for Rubber Property – Resilience by Vertical Rebound

ASTM D5992 - Standard Guide for Dynamic Testing of Vulcanized Rubber and Rubber-Like Materials Using Vibratory Methods

ASTM D7091 - Standard Practice for Nondestructive Measurement of Dry Film Thickness of Nonmagnetic Coatings Applied to Ferrous Metals and Nonmagnetic, Nonconductive Coatings Applied to Non-Ferrous Metals

(Copies of these documents are available online at www.astm.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 10846-1 - Acoustics and Vibration – Laboratory Measurement of Vibro-Acoustic Transfer Properties of Resilient Elements – Part: 1: Principles and Guidelines

ISO 10846-2 - Acoustics and Vibration – Laboratory Measurement of Vibro-Acoustic Transfer Properties of Resilient Elements – Part 2: Direct Method for Determination of the Dynamic Stiffness of Resilient Supports for Translatory Motion

(Copies of these documents are available online at www.iso.org.)

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2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The mounts furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Materials.

3.2.1 Metal. The metal components of the mount shall be manufactured from steel or manganese bronze as specified (see 6.2), in accordance with Drawing 803-1385709. The metal components shall be formed to a shape and finish in accordance with the dimensions and allowable tolerances specified by the drawing.

3.2.2 Rubber. The load-carrying resilient element of the mount shall be a compound using natural rubber (see 6.6.3) as the basic material. The natural rubber components shall be coated with an oil- and ozone-resistant compound (see 6.6.2). The mount components shall be molded to the specified form shown on Drawing 803-1385709. The rubber auxiliary snubber and main snubber components shall be fabricated of an oil- and ozone-resistant compound (see 6.6.3), utilizing polymerized chloroprene or a copolymer product of butadiene and acrylonitrile as the basic material molded to the specified form as shown on Drawing 803-1385709.

3.2.2.1 Porosity and delamination. The rubber elements of the finished mount shall not have any porosity, air pockets, or delamination in any portion when visually examined.

3.2.2.2 Bond. The rubber elements shall be bonded to metal components as specified on Drawing 803-1385709.

3.2.2.3 Surface condition. There shall be no backrinding, blisters, or other defects on the outer surfaces of the rubber elements when examined. Refer to table XI.

3.2.2.4 Physical requirements. Rubber elements shall meet the physical requirements specified in 3.2.2.4.1 through 3.2.2.4.2.

3.2.2.4.1 Rubber stocks. The vulcanized rubber stocks used for the resilient elements, main snubbers, and auxiliary snubbers shall conform to the physical requirements specified in table I. Suggested rubber compounds meeting these requirements are specified in 6.6.3.

TABLE I. Physical requirements for rubber compounds in mount components.

Properties	Resilient element	Main snubbers	Auxiliary snubbers	Test method
Tensile strength (lb/in^2, min.)				
Before aging	2,800	2,300	1,900	4.4.1
After aging at 194 °F for 46 hours	2,200	2,300	1,900	4.4.3
Elongation at break, percent (min.)				
Before aging	500	250	400	4.4.1
After aging at 194 °F for 46 hours	450	180	370	4.4.3
Compression set after oven aging, percent (max.)	35	-	-	4.4.2.1
Cold compression set, percent (max.)	25	-	-	4.4.2.2
Adhesion of resilient element to metal, (lb/in^2, min.)	550	450	450	4.4.6.1

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TABLE I. Physical requirements for rubber compounds in mount components - Continued.

Properties	Resilient element	Main snubbers	Auxiliary snubbers	Test method
Hardness, shore (type A durometer) after 3 seconds	50±5	85±5	60±5	4.4.4
Resistance to oil, increase in volume, percent (max.)	-	10	10	4.4.5.1
Impact resilience, percent (max.)	-	12	9	4.4.10
Specific gravity	3.2.2.4.1.1	3.2.2.4.1.1	3.2.2.4.1.1	4.4.11
NOTE: NAVSEA ltr Ser 61/17-082 may be used as reference for ASTM D2632 test results.				

3.2.2.4.1.1 Specific gravity. The specific gravity of the rubber compounds used for the resilient element, main snubber, and auxiliary snubber shall be determined in accordance with 4.4.11. The specific gravity found during qualification inspection of the contractor's rubber compounds shall be recorded as a requirement with a tolerance of plus or minus 0.02 for subsequent conformance testing of the contractor's rubber compounds.

3.2.2.4.2 Oil- and ozone-resistant coating. The oil- and ozone-resistant coating shall conform to the physical requirements specified in table II. Suggested formulations and application procedures are specified in 6.6.2.

TABLE II. Physical requirements for the oil- and ozone-resistant coating.

Property	Requirement	Test method
Tensile properties of dried film		
Tensile strength (lb/in^2, min.)	2,000	4.4.1
Elongation at break, percent (min.)	400	4.4.1
Properties of coated test specimens		
Resistance to oil, increase in volume, percent (max.)	5	4.4.5.2
Adhesion to resilient element stock	No cracks, breaks, tears, or blisters which allow the coating to be peeled off by hand either before or after immersion in oil.	4.4.6.2
Ozone resistance	No cracks	4.4.7
Flexibility of coating	No cracks	4.4.8
Properties of coated mount		
Appearance	No blisters or other defects	table XI
Film thickness (as measured on coated steel) (min.)	0.003 inch	4.4.9

3.2.3 Protective treatment. The components used in the manufacture of the mount shall be resistant to, or protected against, corrosion by salt water, spray, or other atmospheric conditions encountered in service. Unless otherwise specified (see 6.2), the minimum protective treatment shall be that specified on the mount drawing.

3.2.4 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

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3.2.5 Hazardous materials. Materials and products utilized in this specification should avoid chemicals listed on the NAVSEA List of Targeted Chemicals (N-LTC) contained within T9070-AL-DPC-020/077-2. These chemicals pose significant risk to the user, environment, or both, and are deemed both undesirable and unsustainable by NAVSEA. NAVSEA is minimizing the use of hazardous materials in the design and development of its assets. It is recommended that alternative materials be considered for associated applications, so as to minimize the integration of targeted chemicals in assets called out in this specification.

3.3 Requirements for finished mounts.

3.3.1 Dynamic stiffness. The dynamic stiffness of the mount shall be within the specified limits of table III when loaded axially to 3,500 and 5,000 pounds and tested in accordance with 4.4.13. Unless otherwise specified (see 6.2), the dynamic stiffness test shall be conducted only in the axial direction (see 6.4).

TABLE III. Dynamic stiffness requirements for the mount.

Dynamic stiffness, direction	Axial load, pounds	Calculated resonant frequency, hertz	Dynamic stiffness requirement, lb/in
Axial	3,500	4.0 – 6.0	5,716 – 12,861
	5,000	4.0 – 6.0	8,166 – 18,373
Transverse	3,500	4.0 – 6.0	5,716 – 12,861
	5,000	4.0 – 6.0	8,166 – 18,373
Longitudinal	3,500	1.75 – 3.75	1,093 – 5,020
	5,000	1.75 – 3.75	1,562 – 7,172

3.3.2 Deflection at 5,000 pounds load. The mount shall not deflect more than 0.64 inch when tested as specified in 4.4.12. The mount shall not show any breaks, cracks, tears, or separation between the component parts.

3.3.3 Static load deflection. The mount shall not show any damage, separation, cracks, tears, or breaks between component parts or appreciable permanent deformation of metal parts when tested in accordance with 4.4.14.

3.3.4 Drift. The mount shall not drift more than 0.045 inch under a 5,000-pound load, and the dynamic stiffness shall be within the specified limits of table III when tested in accordance with 4.4.15.

3.3.5 Porosity and delamination. The mount shall not show evidence of porosity in the rubber or separation of the rubber into distinct layers or laminations when tested in accordance with 4.4.16.

3.3.6 Identification. Each mount shall be identified with the markings specified on Drawing 803-1385709.

3.3.7 Mount design. The mount shall conform to all applicable details shown on Drawing 803-1385709. Unless the four-hole configuration is specified (see 6.2), the mount shall be furnished with the six mounting hole configuration in the channel.

3.3.8 Workmanship. The workmanship shall be uniform in quality and the surface shall not contain any foreign matter, corrosion, or imperfections, such as perforations, sharp edges or corners, seams, cracks, laps, dents, raised metal, nicks, scratches, burrs, or other flaws that adversely affect performance. There shall be no defects (see table XI) when examined.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

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4.2 Qualification inspection. Qualification inspection shall consist of the tests specified in 4.2.1 and 4.2.2.

4.2.1 Rubber samples and tests for qualification.

4.2.1.1 Rubber samples. Samples listed in table IV manufactured from each rubber stock specified shall be provided (resilient element, main snubber [decelerator], and auxiliary snubber if applicable [see 6.2]).

TABLE IV. Rubber samples for qualification.

Quantity	Sample	Sample size	Property to be measured
3	Sheet samples of resilient element, decelerator, and auxiliary snubber stock	0.08±0.01 x 6 x 6 inches	Tensile strength, ultimate elongation, specific gravity
6	Cylinder or disc of resilient element stock	0.50±0.02 inch thick x 1.14±0.02-inch diameter	Compression set, hardness
3	Cylinder or disc of decelerator and auxiliary snubber stock	0.50±0.02 inch thick x 1.14±0.02-inch diameter	Hardness, resilience
3	Metal adhesion samples of resilient element, decelerator, and auxiliary snubber stock	In accordance with ASTM D429, Method A	Rubber-to-metal adhesion
3	Sheet samples of decelerator and auxiliary snubber stock	0.08±0.01 x 1 x 2 inches	Resistance to oil
6	Sheet samples of resilient element stock with oil and ozone resistant coating	0.08±0.01 x 1 x 2 inches	Resistance to oil (3 samples), adhesion to resilient element stock (3 samples)
3	Sheet samples of resilient element stock covered with oil- and ozone-resistant coating	0.08±0.01 x 1 x 6 inches	Ozone resistance (2 samples), flexibility of coating (1 sample)
1	Sample of oil and ozone resistant coating	0.006±0.003 x 6 x 6 inches	Tensile strength, ultimate elongation

4.2.1.2 Uncoated rubber samples. The uncoated rubber stock samples specified in table IV shall meet the requirements of table I when subjected to the tests specified in table V.

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TABLE V. Tests on uncoated rubber samples.

Test	Inspection	Requirement
Initial tensile strength	4.4.1	table I
Initial ultimate elongation	4.4.1	table I
Tensile strength after oven aging	4.4.1 and 4.4.3	table I
Ultimate elongation after oven aging	4.4.1 and 4.4.3	table I
Compression set after oven aging	4.4.2.1 and 4.4.3	table I
Cold compression set	4.4.2.2	table I
Resistance to oil	4.4.5.1	table I
Adhesion to metal	4.4.6.1	table I
Impact resilience	4.4.10	table I
Hardness	4.4.4	table I
Specific gravity	4.4.11	table I

4.2.1.3 Oil and ozone resistant coating samples. The oil- and ozone-resistant coating and coated samples specified in table IV shall meet the physical requirements listed in table II when subjected to the tests specified in table VI.

TABLE VI. Tests on coated rubber samples and dried film.

Test	Inspection	Requirement
Tensile strength of dried film	4.4.1	table II
Elongation of dried film	4.4.1	table II
Resistance to oil	4.4.5.2	table II
Adhesion to resilient element stock	4.4.6.2	table II
Ozone resistance	4.4.7	table II
Flexibility	4.4.8	table II

4.2.2 Qualification inspection on finished mounts.

4.2.2.1 Mount samples. If the rubber samples specified in 4.2.1.1 meet all requirements in 4.2.1, the tests specified below shall be conducted using four complete mounts (including snubber components) manufactured from the respective rubber stock.

4.2.2.2 Group one. Two of the four mounts specified in 4.2.2.1 shall be subjected to the examination and tests in the sequence shown in table VII.

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TABLE VII. Test group one.

Test	Inspection	Requirement
Examination	table XI	No defects
Deflection at 5,000 pounds load	4.4.12	3.3.2
Dynamic stiffness	4.4.13	3.3.1
Static load-deflection	4.4.14	3.3.3
Film thickness of oil- and ozone-resistant coating	4.4.9	table II

4.2.2.3 Group two. The remaining two mounts shall be subjected to the examination and tests in the sequence shown in table VIII.

TABLE VIII. Test group two.

Test	Inspection	Requirement
Examination	table XI	No defects
Deflection at 5,000 pounds load	4.4.12	3.3.2
Dynamic stiffness	4.4.13	3.3.1
Drift	4.4.15	3.3.4
Porosity and delamination	4.4.16	3.3.5

4.3 Conformance inspection. Conformance inspection shall be performed on each production lot and includes the examination and tests contained in 4.3.2 and 4.3.3.

4.3.1 Lot. For the purpose of conformance inspection and sampling, a lot shall consist of all the mounts of one type, design, and load rating offered for delivery. A lot shall not exceed 300 mounts. A lot serial number shall be assigned to each mount. The lot serial number shall not be repeated in any one quarter. The serial number shall be traceable to the rubber batch numbers, the manufacturing and process records against which the mounts and snubbers were manufactured, and all conformance requirements as required by this specification.

4.3.2 Conformance tests on rubber stock. Samples of rubber stock shall meet the physical requirements listed in table I when subjected to the tests specified in table IX. The samples shall be certified to be of the same material and equivalent cure as the corresponding components in the lot of finished mounts offered for delivery. If any samples tested fail to meet the requirements, all mounts in the lot represented by the samples shall be rejected.

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TABLE IX. Samples and tests on rubber stock.

Quantity	Sample	Sample size	Property	Test
2	Sheet samples of resilient element, decelerator, and auxiliary snubber stock	0.08±0.010 x 6 x 6 inches	Tensile strength, ultimate elongation, specific gravity	4.4.1 4.4.11
1	Cylinder or disc of resilient element, decelerator, and auxiliary snubber stock	0.5±0.02 inch thick x 1.14±0.02-inch diameter	Hardness	4.4.4
1	Cylinder or disc of resilient element stock	0.5±0.02 inch thick x 1.14±0.02-inch diameter	Compression set	4.4.2.1

4.3.3 Conformance inspection on mounts. Unless otherwise specified (see 6.2), production mounts (including snubber components) from each lot shall be inspected as specified in 4.3.3.1 through 4.3.3.5.

4.3.3.1 Sampling for visual and dimensional examination. As a minimum, a random sample of mounts from a lot shall be inspected in accordance with the sampling plan specified herein. MIL-STD-407 shall be used to determine and evaluate defects through visual examination. The classification and types of defects are shown in table XI. The sample size is shown in table X. If one or more defects are found in any sample, the defective parts shall be rejected and the entire lot screened for all defects. All defective parts shall be rejected and removed from the lot.

TABLE X. Sample size for visual inspection.

Number of mounts in a lot	Number of mounts in sample
90 and under	8
91 to 200	20
201 to 300	30

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TABLE XI. Classification of physical defects.

Categories	Defects
Major	
101	Evidence of use of unauthorized materials.
102	Rubber elements not molded to specified form on drawing.
103	Evidence of delamination or air pockets in rubber elements of finished product.
104	Rubber elements not bonded to steel components in conformance with drawing.
105	Evidence of backrinding, blisters, cracks, breaks, gouges, tears, or other imperfections on outer surface of rubber elements that can affect performance or service life.
106	Evidence of tackiness, non-drying, peeling, or other imperfections in the oil-resistant coating. Required coating absent.
107	Steel components not protected from corrosion by salt water or spray or other atmospheric conditions encountered in service.
108	Dimensions, configuration, and construction not in conformance with drawing.
109	Any other defect that would affect the serviceability, performance, or service life of the mounts.
Minor	
201	Identification marking not distinct.
202	Burrs, rough edges, and sharp corners not removed.
203	Any other defect that would not affect the serviceability, performance, or service life of the mounts.

4.3.3.2 Conformance test to be performed on all mounts in the lot. Prior to sampling for conformance tests, all mounts in the lot shall meet the requirement for deflection at 5,000 pounds (see 3.3.2) and shall not exhibit any defects when tested in accordance 4.4.12. All mounts that do not meet this requirement shall be rejected, removed from the lot, and shall not be used for further conformance testing.

4.3.3.3 Sampling for tests. Unless otherwise specified (see 6.2), conformance tests shall be performed on a random sample of mounts from a lot in accordance with the sampling plan specified in table XII.

TABLE XII. Sampling for conformance tests.

Number of mounts in a lot	Number of mounts in sample
3 to 15	3
16 to 20	4
21 to 25	5
26 to 90	8
91 to 150	13
151 to 280	20
281 to 300	32

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4.3.3.4 Conformance tests to be performed on mount samples. The mounts selected in accordance with 4.3.3.3 shall be subjected to the tests specified in table XIII in the order listed. If porosity-delamination requirements have not been met by all mounts in the sample, the entire lot shall be rejected. If static load deflection requirements have not been met by all mounts in the sample, the entire lot shall be rejected. If dynamic stiffness, examination, and film thickness requirements have not been met by all mounts in the sample, the defective parts shall be rejected and the entire lot shall be tested for the defect(s) found. All defective parts shall be rejected and removed from the lot.

TABLE XIII. Conformance tests on mount samples.

Test	Requirement	Inspection
Dynamic stiffness	3.3.1	4.4.13
Static load-deflection	3.3.3	4.4.14.2.2
Film thickness of oil- and ozone-resistant coating	table II	4.4.9
Porosity and delamination [1/]	3.3.5	4.4.16
Examination	No defects	table XI

NOTE:
[1/] Only one sample per lot shall be subjected to this test.

4.3.3.5 Conforming mounts. Every mount that passes the requirements for conformance inspection, except the one that is destroyed for the porosity and delamination test, can be returned to the lot.

4.4 Method of tests. Tests shall be performed within an ambient temperature range of 75±5 °F.

4.4.1 Tensile strength and ultimate elongation tests. Tensile strength and ultimate elongation tests shall be conducted in accordance with ASTM D412, Method A. Standard Die C shall be used for cutting dumbbell specimens.

4.4.2 Compression set. Compression set tests shall be conducted as specified in 4.4.2.1 and 4.4.2.2.

4.4.2.1 Compression set after oven aging. Compression set shall be determined in accordance with Method B of ASTM D395, except oven aging shall be as specified in 4.4.3.

4.4.2.2 Cold compression set. Cold compression set shall be determined in accordance with ASTM D1229, except the exposure shall be at 30±2 °F for 94±½ hours. Compression set shall be determined 30 minutes after release from compression.

4.4.3 Oven aging test. The specimens for tensile, ultimate elongation, and compression set tests shall be oven aged in accordance with ASTM D573 at a temperature of 194±2 °F for 46±¼ hours. Final determination of aged tensile and ultimate elongation specimens shall be made not less than 16 hours or more than 48 hours after removal from the oven. Tensile and ultimate elongation tests on unaged specimens shall be made immediately prior to, and on the same machine as, the tensile tests on the oven aged specimens (see 4.4.1).

4.4.4 Hardness test. The hardness of the 0.50-inch thick rubber specimens shall be determined in accordance with ASTM D2240 using a type A shore durometer. A 3-second reading shall be taken.

4.4.5 Resistance to oil. Tests for resistance to oil shall be conducted as specified in 4.4.5.1 and 4.4.5.2.

4.4.5.1 Uncoated specimen. The increase in volume of the rubber specimens due to immersion in oil shall be determined in accordance with ASTM D471, except the thickness of the specimens shall be 0.08±0.01 inch. The specimens shall be immersed in IRM 903 reference oil as specified in ASTM D471 for a period of 46¼ hours at 73±2 °F.

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4.4.5.2 Coated specimen. The increase in volume of the coated specimens due to immersion in oil shall be determined in accordance with ASTM D471, except the coated specimens shall be as specified in table IV and the immersion period shall be 70±¼ hours at 158±2 °F. The specimens shall be immersed in IRM 903 reference oil as specified in ASTM D471.

4.4.6 Adhesion tests. Adhesion tests shall be conducted as specified in 4.4.6.1 and 4.4.6.2.

4.4.6.1 Adhesion to metal. The tests for adhesion to metal shall be in accordance with Method A of ASTM D429. Three specimens of each rubber stock shall be tested and the results averaged for each stock.

4.4.6.2 Adhesion of oil- and ozone-resistant coating. The adhesion of the oil and ozone resistant coating to three 0.08- by 1- by 2-inch samples of the resilient rubber specimens (see table IV) shall be determined before and after immersion in oil (see 4.4.5.2). Each coated specimen shall be flexed, elongated by hand, and then visually examined for adhesion failures. The coated specimens shall not exhibit cracks, breaks, tears, blisters, or other defects conducive to peeling of the coating by hand, either before or after immersion in oil.

4.4.7 Ozone resistance. The ozone resistance of the coated resilient components of the mounts shall be determined on two coated 1- by 6- by 0.08-inch thick specimens (see table IV). The specimens shall be elongated to 20 percent extension. A thin layer of melted paraffin wax shall be applied to each of the four surfaces of the stretched specimens in an area where it is clamped, but not exceeding a ¼-inch width from the clamps. The stretched specimens shall be conditioned for 16±2 hours at 104±4 °F before being exposed in accordance with ASTM D1149. The standard ozone partial pressure shall be 100 millipascals (mPa) ±10 percent, the temperature shall be 104±4 °F, and the period of exposure shall be 168 hours. Observation magnification shall be 7X.

4.4.8 Flexibility of the coating. The flexibility of a coated 1- by 6- by 0.08-inch resilient element rubber specimen specified in table IV shall be determined. Each specimen shall be clamped in a universal test machine and a tensile force applied to elongate the specimen 100 percent at a loading rate of 20 inches per minute. The force shall then be released so that the specimen is no longer elongated. The cycle shall be repeated six times. The coated specimens shall be visually examined and shall exhibit no cracks, breaks, tears, or blisters.

4.4.9 Film thickness of the oil- and ozone-resistant coating. The film thickness of the oil- and ozone–resistant coating shall be measured on the coated metal steel plate to determine conformance with the requirement of table II. The thickness of the coating shall be determined using either a magnetic type thickness gauge or a micrometer to measure the thickness of the steel plate before and after the coating is applied, or by other valid and accurate procedures. Refer to ASTM D1005 and ASTM D7091. (The coating thickness shall not be determined by measuring the thickness of different coated and uncoated steel plates, unless the procedure is demonstrated to be valid and accurate.)

4.4.10 Impact resilience test. Impact resilience tests shall be in accordance with ASTM D2632. Three specimens shall be tested and the results averaged.

4.4.11 Specific gravity. The specific gravity of the rubber samples shall be determined according to ASTM D792 on specimens cut from 0.08- by 6- by 6-inch sheets. The average of three measurements shall be used to determine conformance with 3.2.2.4.1.1.

4.4.12 Test for deflection at 5,000 pounds load. A universal type testing machine shall be used. Data shall be corrected for error resulting from flexibility in the test apparatus. A mount without snubber cones shall be secured on a jig. It shall be subjected to four loading cycles in the axial direction (see 6.4). Three loadings shall be at a rate not greater than 0.3 inch per minute. The fourth loading cycle shall be at a rate of 0.05 inch per minute. The deflection at 5,000 pounds load for the fourth loading cycle shall be measured to the closest 0.001 inch to determine conformance. While deflected, the mount shall be examined for a break or separation between the component parts.

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4.4.13 Dynamic stiffness test. Unless otherwise specified (see 6.2), dynamic stiffness of the mount shall be measured in the axial direction at test loads of 3,500 and 5,000 pounds to determine conformance with 3.3.1. When dynamic stiffness measurements in the transverse and longitudinal directions are required (see 6.2), a fixture similar to that shown on figure 1 shall be used to determine if the per mount dynamic stiffness is in compliance with 3.3.1. Prior to conducting dynamic stiffness tests, all mounts shall first be "conditioned" a minimum of three cycles to the maximum test load. The general guidelines and considerations for dynamic stiffness testing given in ASTM D5992, ISO 10846-1, and ISO 10846-2 shall be followed.

4.4.13.1 Test system for measuring dynamic stiffness. The test system shall be capable of superimposing cyclical vibration while statically loading the mount. Refer to ASTM D5992 for examples of non-resonant forced vibration test systems and guidance in addressing mechanical and instrumentation factors that influence dynamic measurements. A commonly used type of equipment for non-resonant testing is a closed-loop servo-hydraulic machine (see figure 2). Transducers shall be arranged to measure force at the blocked output or foundation side of the mount and displacement at the equipment or input side of the mount. The stiffness of the test apparatus (machine, load-frame, force transducer, etc.) in series with the mount under test shall be determined experimentally (see 6.5) or via calculation. A test apparatus that is at least 100 times stiffer than the dynamic stiffness of the mount being tested shall not require measurement adjustment to account for the flexibility of the test apparatus. If the test apparatus fails to meet this relative stiffness requirement, the test apparatus compliance shall be subtracted from the measurement to obtain the dynamic stiffness of the mount (K_{mount}). The dynamic stiffness of the mount can be calculated using the classic equation governing the summation of springs in series:

$$(1/K_{measured}) = (1/K_{mount}) + (1/K_{apparatus}) \qquad \text{yielding} \qquad K_{mount} = \frac{(K_{apparatus})(K_{measured})}{(K_{apparatus}) - (K_{measured})}$$

To avoid over-correction, the stiffness of the test apparatus shall be greater than 30 times the stiffness of the mount.

4.4.13.2 Confirmation of the measurement system. Refer to ASTM D5992 for mechanical and instrumentation factors influencing dynamic measurement and selection and use of springs to assess the capability of the test apparatus to measure dynamic stiffness. Prior to conducting dynamic stiffness measurements on mounts, the dynamic calibration of the complete system shall be confirmed using steel springs of known spring constants, having negligible damping. Springs shall be free of hysteresis and their dynamic and static stiffness (tangent to the slope of the load-deflection curve) shall be equivalent. This procedure is not intended to replace the periodic calibration of transducers, signal conditioners, and other measuring instruments. Springs shall be selected on the basis of their linear force-deflection properties and ability to produce a spring constant comparable to the stiffness of the mount to be tested. Refer to ASTM D5992 for mechanical and instrumentation factors influencing dynamic measurement and selection of springs. System confirmation shall be performed using springs of known spring constants at the excitation amplitudes and frequencies at which the mount is to be tested. System confirmation shall be performed at loads that produce spring constant measurements that bracket the expected dynamic stiffness range of the mount to be tested. This shall be accomplished by adding or removing springs within the stiffness range of intended mount measurements. Care shall be taken to ensure measurements are conducted within the linear load range of the springs. All spring dynamic stiffness measurements (corrected for test apparatus stiffness) should be within 2 percent of the known spring constants. If this is not the case, the error shall be investigated and corrected. A system scaling factor may be applied to measured dynamic stiffness with NAVSEA approval. Spring constant values shall be verified by NAVSEA during qualification and when requested by the mount manufacturer.

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4.4.13.3 Measuring mount dynamic stiffness. Unless otherwise specified (see 6.2), dynamic stiffness shall be measured on the mount in the axial direction at loads of 3,500 and 5,000 pounds to determine conformance with 3.3.1. When dynamic stiffness in the transverse and longitudinal directions is required, a fixture similar to that shown on figure 1 shall be used to determine if the per-mount dynamic stiffness is in compliance with 3.3.1. Sinusoidal excitation displacement amplitude shall be approximately 0.020 inch peak-to-peak at 5 hertz for dynamic stiffness testing in the axial and transverse directions, and at 2.75 hertz for testing in the longitudinal direction. Dynamic stiffness measurements shall be acquired at the excitation frequency by using Fast Fourier Transform (FFT) based methods. This requires measurement of the sinusoidal displacement at the input of the mount and transmitted dynamic force at the blocked output of the mount, as well as the phase relationship between them. It should be noted that damping loss factor is the tangent of the phase between the displacement and force. The dynamic stiffness is calculated from the measurements as follows:

$$K_{measured} = (F/X)(\cos(\theta))$$

Where: (F) and (X) are the amplitudes of sinusoidal force and displacement respectively and (θ) is the phase between them. Loss factor (η) can be calculated from the measurements as follows.

$$\eta = \tan(\theta)$$

In order to obtain mount stiffness (K_{mount}) to determine compliance with 3.3.1, the measured stiffness ($K_{measured}$) may likely require adjustment to account for test apparatus flexibility (see 4.4.13.1).

$$K_{mount} = \frac{(K_{apparatus})(K_{measured})}{(K_{apparatus}) - (K_{measured})}$$

A system scaling or multiplication factor (C) applied to the measured dynamic stiffness ($C*K_{measured}$) is assumed to be unity (C=1) and not shown in the above equation.

4.4.14 Static load-deflection tests. The mount's static load-deflection shall be determined as specified in 4.4.14.1 through 4.4.14.3. Main snubbers shall be in place, but auxiliary snubbers shall not be used. Data shall be corrected for error resulting from flexibility in the test apparatus.

4.4.14.1 Apparatus. A universal type testing machine shall be employed to conduct static load-deflection tests on the mounts. The testing machine shall be equipped with a control that will load the mount at a constant deflection rate.

4.4.14.2 Test in axial direction. Testing in the axial direction (see 6.4) on the universal testing machine shall be as specified in 4.4.14.2.1 and 4.4.14.2.2.

4.4.14.2.1 Qualification load-deflection testing. When load-deflection tests are conducted in the axial direction, a single mount shall be secured in a jig and subjected to four loading and unloading cycles. For the first three loading cycles, the loads shall be applied at a deflection rate not exceeding 0.3 inch per minute up to 50,000 pounds. The fourth loading cycle shall be conducted at a rate of 0.05 inch per minute, and the mount shall be loaded up to 70,000 pounds. The fourth loading and unloading cycle shall be reported as the static load-deflection curve for the mount.

4.4.14.2.2 Conformance load-deflection testing. The load-deflection test shall be as specified in 4.4.14.2.1, except that the mounts shall be subjected to four loading and unloading cycles in the axial direction as specified herein. For each loading cycle, loads shall be applied up to 50,000 pounds. The deflection rate shall not exceed 0.3 inch per minute. The mount shall not show any damage or permanent deformation.

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4.4.14.3 Test in transverse and longitudinal directions. When load-deflection tests are conducted in a transverse or longitudinal direction (see 6.4), two mounts shall be secured in a jig similar to that shown on figure 1. Each mount shall be compressed axially to the amount that its upper load deflected the mount during the static load-deflection test in the axial direction. This compression is obtained by adjusting the nuts on the four rods, which changes the distance between the channels of the mounts with respect to the plane of loading bar. After the two mounts have been assembled in the jig and secured, they shall be subjected to four loading and unloading cycles to 50,000 pounds (25,000 pounds per mount) in accordance with the procedures specified in 4.4.14.2.

4.4.15 Drift test. The bottom portion of the rubber resilient element shall not touch the main snubbers during the drift test. Auxiliary snubbers shall not be installed. The unloaded mount height shall be measured from an arbitrarily selected datum line, usually the base or top side of the channel web. A load of 5,000 pounds shall then be applied in the axial (vertical) direction on the mount. One hour after applying the load, the deflection of the mount from the no-load position shall be measured in mils. Thereafter, drift and ambient air temperature measurements shall be made every hour for 6 hours followed by measurements at 24-hour intervals for 10 days. Temperatures shall not vary by more than ±2 °F during the drift test. The drift of the mount shall be the difference in deflection taken 1 hour after loading and 10 days after loading. Within 15 minutes after the final reading has been obtained, the mount dynamic stiffness shall be measured under a load of 5,000 pounds (see 4.4.13). The load shall then be removed and the mount height allowed to recover. After complete recovery, the unloaded height of the mount shall be measured.

4.4.16 Porosity and delamination. From each lot, one of the mounts previously used for static load-deflection tests shall be cut into two parts as shown on figure 3. The cut surfaces of the rubber shall be carefully examined for porosity. The parts shall be immersed in toluene for 24 hours. After removal from toluene, the rubber parts shall be examined for evidence of separation into distinct layers or laminations.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The mount covered in this specification is intended to support loads only in the axial direction for noise and vibration attenuation. The mount is primarily for use in naval submarines, and in naval surface ships, where applicable. Current stock of the mount may be used until depleted; future requisitions for the mount should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Whether metal parts are to be manufactured from steel or manganese bronze (see 3.2.1).

c. Protective treatment, if different from drawing (see 3.2.3).

d. Whether dynamic stiffness is to be measured in a direction other than just axial (see 3.3.1, 4.4.13, 4.4.13.3, and 6.4).

e. Whether four-hole configuration is required (see 3.3.7).

f. Whether auxiliary snubbers are required (see 4.2.1.1 and 6.4).

g. If necessary, that a sampling plan for conformance inspection, including acceptance and rejection levels, is to be provided (see 4.3.3 and 4.3.3.3).

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h. Packaging requirements (see 5.1).

i. Marking and special marking requirements (see 6.8).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of the award of contract, qualified for inclusion in Qualified Products List QPL No. 19863 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and the manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.3.1 Additional qualification testing provisions. NAVSEA reserves the right to witness tests and perform any of the tests set forth herein where such testing is deemed necessary to assure compliance to prescribed requirements of the qualification tests. Qualification testing may be conducted only after NAVSEA has authorized such testing in writing. Copies of SD-6, "Provisions Governing Qualification," are available online at http://quicksearch.dla.mil.

6.4 Definitions.

6.4.1 Auxiliary snubbers. Snubbers used to limit shock excursions in the downward direction (see 6.2).

6.4.2 Axial (vertical) direction. A direction that is perpendicular to the top steel plate (see figure 3).

6.4.3 Transverse direction. A direction that is perpendicular to the flanges of the steel support channel (see figure 3).

6.4.4 Longitudinal direction. A direction that is parallel to the flange of the steel support channel (see figure 3).

6.5 Measurement of test apparatus stiffness. Where practicable, the stiffness of those components of the test apparatus in series with the specimen should be determined experimentally as follows:

a. For test machines capable of such operation, place the machine in the load feedback mode with the displacement transducer on its most sensitive range and the force transducer on its full scale setting.

b. Bring the platens of the machine either directly in contact or against a stiff steel spacer separating the platens. **Caution: This procedure is not recommended for test machines not equipped with a safety interlock system to prevent the force from exceeding a predetermined value.**

c. The force applied should be sufficient to ensure good resolution from both the force and the displacement instrumentation.

d. Sufficient points should be taken to ensure that the force-deflection relationship is linear and that effects such as non-parallelism between platens do not produce nonlinear results.

e. The stiffness of the test apparatus $K_{apparatus}$ should be determined from the "best fit" straight line through the data points of the force-deflection curve, disregarding anomalous (i.e., abnormal or irregular) points that may occur at low loads prior to proper seating of the test machine components.

f. Where it is not practicable to experimentally determine the stiffness of the test apparatus, the manufacturer's published stiffness for each component in series with the test specimen should be used to calculate the overall stiffness of the test apparatus.

6.6 Suggested formulations and application procedure.

6.6.1 Disclaimer. The furnishing of information, formulations, and procedures (see 6.6), are solely for assistance in fabrication of the 5B5,000H mount for naval use. The use of this information does not constitute any agreement or obligation by the Department of the Navy to procure mounts made of these formulations. Also, the use of this information does not guarantee compliance with this specification nor will it relieve the manufacturer from having the mounts tested against the applicable test requirements of this specification.

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6.6.2 Oil- and ozone-resistant coating.

6.6.2.1 Formulations for coating. The formulations shown below should be blended together just prior to use. The viscosity of the blend should be adjusted to about 650 centipoises by using more or less of methyl ethyl ketone in the formulations.

TABLE XIV. Suggested coating formulations.

	312-118A	**312-118B**
Prefluxed blend of 70 percent nitrile rubber with 30 percent polyvinyl chloride	100	100
High Abrasion Furnace (HAF) carbon black	30	30
Stearic acid	1	1
Zinc oxide	5	5
ZBX (zinc butyl xanthate)	8	--
DBA (dibenzyl amine)	--	8
Sulfur	--	4
Methyl ethyl ketone	656	674

6.6.2.2 Procedure for application of coating to mounts. The sequence of operations shown below should be followed in order to obtain satisfactory performance from the coating.

a. Remove rubber flash from mount by trimming or buffing.
b. Sandblast the mount, and then brush off dust.
c. Apply primer, in accordance with manufacturer's recommendation, to all metal surfaces.
d. Wash the rubber portion of the mount with an aqueous solution of Vel detergent, or equal, and rinse with tap water.
e. Chlorinate the rubber surfaces by immersing entire mount for 3 minutes in water saturated with chlorine.
f. Rinse mount with tap water and dry until all traces of water have evaporated.
g. Blend the 312-118A solution with the 312-118B solution and let stand 1 hour to permit entrapped air to escape.
h. Apply two coats of the blended formulations to the entire mount, including metal parts, by brushing or dipping.
i. Allow the first coat to dry for at least 1 hour and the second coat to dry for at least 4 days before handling the mount. Alternatively, the second coat should be allowed to dry for at least 4 hours, and then the mount placed in an oven at 130±5 °F for 16 hours to cure the coating.

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6.6.3 Formulations for rubber stock. Table XV contains suggested rubber stock formulations.

TABLE XV. Suggested rubber stock formulations.

Resilient element		Main snubber		Auxiliary snubber		Bonding cement	
Natural rubber	100	Butadiene-high acrylonitrile copolymer	100	Butadiene-high acrylonitrile copolymer	100	Two step phenolic resin	100
N990 carbon black	50	SRF carbon black	75	SRF carbon black	75	Methyl ethyl ketone	150
Stearic acid	3	Stearic acid	1	Stearic acid	1	Methanol	150
Zinc oxide	5	Zinc oxide	5	Zinc oxide	5		
Phenyl-alpha-naphthylamine	1	Two step phenolic resin	30	N,N'-bis-(1 ethyl-3-methyl pentyl)-p-phenylenediamine	2		
2,2'-methylene-bis-(4-methyl-6-t-butylphenol)	1	Coumarone-indene resin	20	Blend of hydrocarbon waxes	3		
ASTM type 101 petroleum oil	3	N,N'-bis-(1 ethyl-3-methyl pentyl)-p-phenylenediamine	2	Coumarone-indene resin	30		
Benzothiazyl disulfide	1	Blend of hydrocarbon waxes	3	Benzothiazyl disulfide	1.5		
Tetramethyl thiuram monosulfide	0.3	Benzothiazyl disulfide	1.5	Sulfur	1.5		
Sulfur	1.5	Salicylic acid	0.5				
		Sulfur	1.5				

NOTE:
1. Rubber formulations based on parts per hundred rubber (phr).

6.7 Marking. Waterproof marking should be provided on interior packages and exterior shipping containers with the following information (see 6.2):

MOUNT TYPE:

MILITARY SPEC NO: MIL-DTL-19863

DATE OF MFG:

LOT NO:

SHELF LIFE: 7 YEARS FROM DATE OF MFG

Note: The date of manufacture should correspond to the elastomer cure date.

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6.8 Subject term (key word) listing.

Decelerator

Noise, structureborne

Rubber stocks

Shock

Snubber

Vibration

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

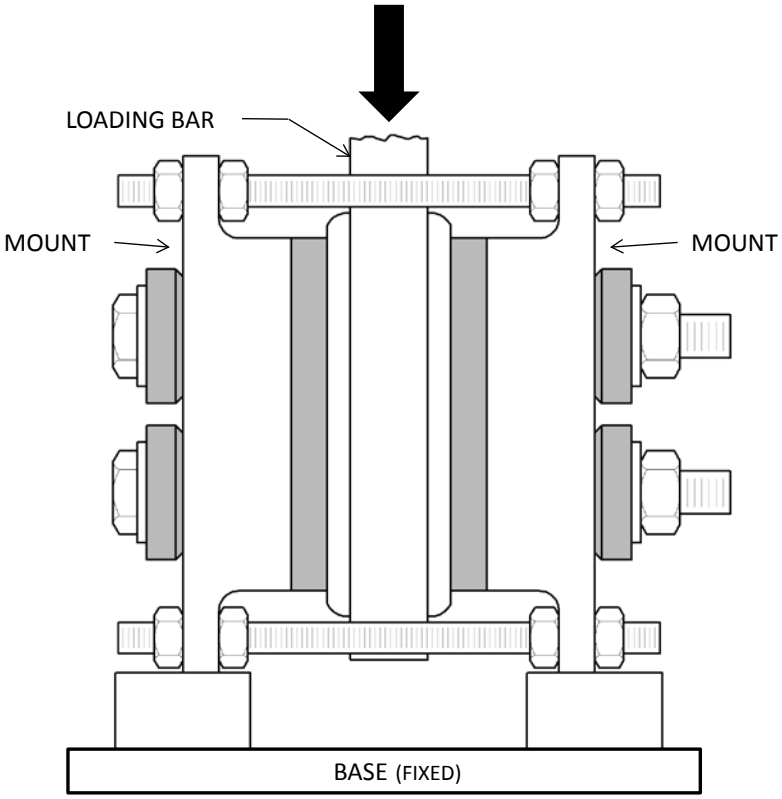


FIGURE 1. Suggested fixture for static load deflection tests in transverse and longitudinal direction.

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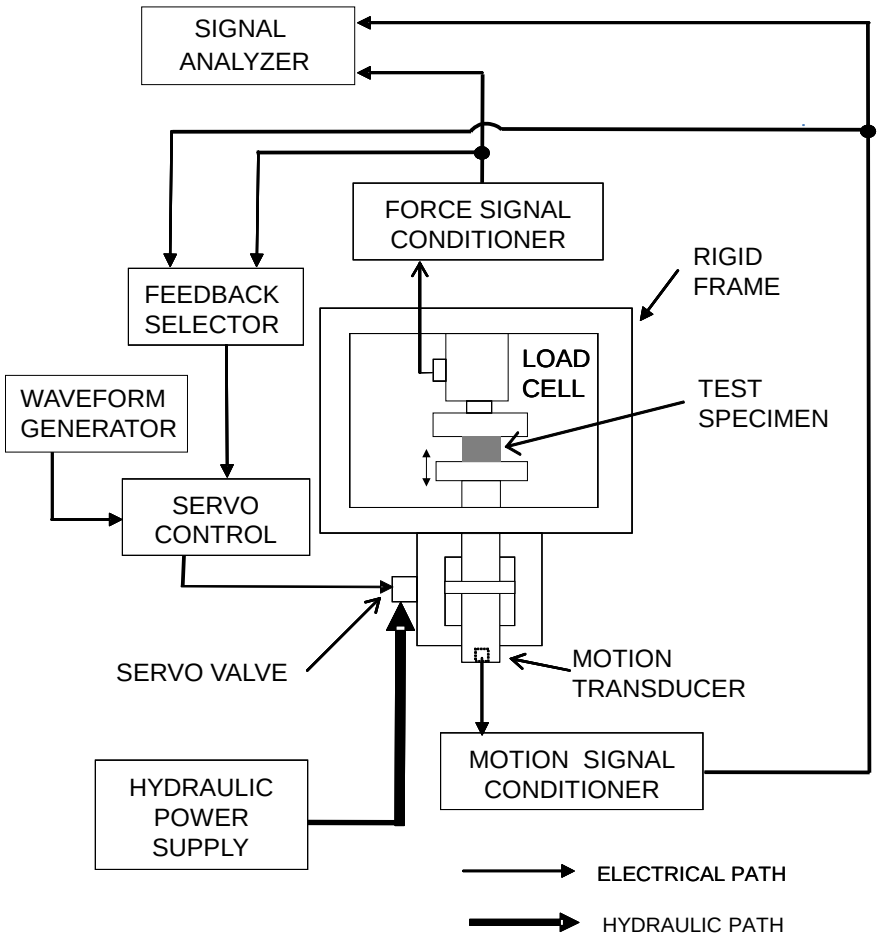


FIGURE 2. Schematic diagram of major components of electrohydraulic closed-loop test machine.

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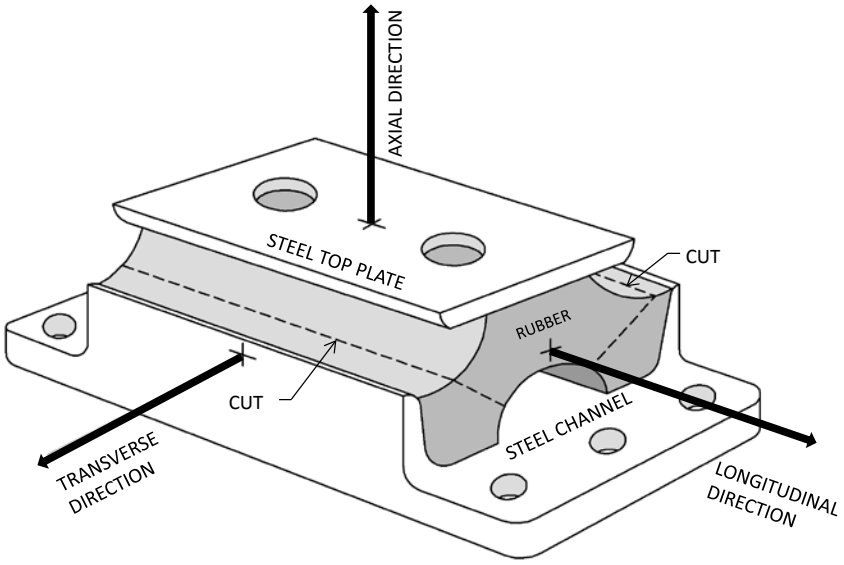


FIGURE 3. Sketch showing axes of type 5B5,000H resilient mount and location of cuts for porosity and delamination test.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 5342-2016-003)

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